Merchant DNA

Cross-merchant fraud intelligence for payment platforms — behavioral scoring and network analysis that catch what single-merchant tools structurally can't see.

Date: September 04, 2026	Scope: Cross-Merchant Risk Intelligence	Target Domain: Payment Aggregators & FinTechs

The Blind Spot

Traditional payment fraud detection operates at the individual transaction level, evaluating buyer card velocity, 3DS challenges, and IP geolocation in isolation. However, modern merchant-side fraud operates through distributed networks:

- Distributed Syndicate Shells: Fraud rings register 10 to 50 seemingly legitimate storefronts with valid KYC documents. Each merchant maintains low, steady volumes that stay well below transaction-level velocity tripwires.
- Mule Accounts & Rapid Cashouts: Sleeper accounts remain dormant for 30–60 days to establish baseline history, then process concentrated transaction bursts followed by immediate settlement withdrawal requests before chargebacks arrive.
- Aggregator Liability: When merchant defaults and chargebacks occur, payment aggregators bear financial liability for unrecoverable negative balances.

The Approach

Merchant DNA addresses cross-merchant risk through three interconnected components:

1. Behavioral Telemetry (60% Weight): A gradient boosted decision tree model evaluating 30+ merchant-level features, including category ticket deviations, off-peak velocity, buyer concentration (HHI), and settlement turnaround speed.
2. Network Graph Analysis (40% Weight): Bipartite graph modeling and Louvain community detection to uncover shared devices, phone numbers, bank accounts, and UPI VPAs.
3. Grounded Synthesis & Bounded Actions: An AI copilot providing grounded synthesis constrained to structured evidence inputs, coupled with reversible settlement holds that protect liquidity while preserving checkout operations for review.

How It Works

- Step 1 — Entity Ingestion & Indexing: Transaction streams, onboarding metadata, and settlement requests are indexed into reverse lookup tables linking merchants to physical and digital identifiers.
- Step 2 — Behavioral Feature Extraction: Rolling windows compute catalog baseline divergence, off-peak processing ratios, round-denomination frequencies, and payout turnaround velocity.
- Step 3 — Graph Topology & Community Detection: Multi-entity bipartite projection generates merchant-to-merchant relationship graphs, applying community clustering to identify syndicates.
- Step 4 — Hybrid Calibrated Risk Scoring: Combines behavioral ML probabilities (60%) and graph density/centrality scores (40%) into a 0–100 risk score with calibrated tiers (Low, Medium, High, Critical).
- Step 5 — Forensic Investigation & Grounded Synthesis: Analysts inspect cases across four core questions (Who, How, Who Else, What Next) supported by an interactive Cytoscape network graph and evidence-constrained AI summaries.

- Step 6 — Bounded Actions & Immutable Audit: Analysts apply targeted holds or clear cases, with all state transitions recorded in an append-only audit ledger.

Stack

Layer	Technologies	Purpose & Role in Pipeline
Frontend	React 18, TypeScript, Tailwind CSS, Vite, Lucide Icons, Recharts	Enterprise analyst console with sub-second page transitions, interactive triage queues, and forensic views.
Graph Visualization	Cytoscape.js, Cola Physics Layout	Multi-entity ego graph visualizer with node clustering and shared-identifier inspection.
Backend Services	FastAPI, Python 3.14, Pydantic v2, Uvicorn, AnyIO	Asynchronous REST API gateway with schema validation, CORS security, and modular service routing.
Machine Learning & Graph	Scikit-Learn (HistGradientBoosting), NetworkX, NumPy, SciPy	Behavioral gradient boosting classifier, 30+ feature extractors, and Louvain community detection.
AI Synthesis	Google Gemini API / Deterministic Grounded Fallback	Grounded synthesis constrained to structured evidence inputs, providing factual case briefings.
Testing & Quality	Pytest, Oxlint, Vite Production Compiler	Automated unit tests, integration suites, API contract validation, and data leakage prevention tests.

Where This Applies

- Payment Aggregators (Razorpay, Stripe, Cashfree): Gatekeeping automated daily settlement batches to intercept coordinated syndicates before payout disbursement.
- Neobanks & Instant Payout Platforms (RazorpayX): Real-time risk scoring on instant settlement drawdowns, protecting automated liquidity pools from compromised credentials.
- Acquiring Banks & Card Networks (RuPay, Visa, Mastercard): Portfolio-level merchant risk surveillance and syndicated chargeback liability mitigation.
- E-Commerce Marketplaces: Detecting seller collusion, device-farm review manipulation, and bust-out accounts.

Business Model

All five tiers run on the same underlying risk and graph intelligence engine — the free tier is not a cost center, but an essential network asset that strengthens the cross-merchant ring-detection signals that paid tiers depend on.

Tier	Target Segment	Key Features	Commercial Model
1. Starter	Individual & informal sellers (WhatsApp/Instagram, gig sellers)	Public trust badge only. Serves as a funnel/network-effect tier (more merchants on platform strengthens cross-merchant ring detection for every paying tier).	Free or nominal ■49–■99/mo
2. Growing SMB	Small D2C brands, single-location retailers	Basic risk dashboard, public trust badge, limited fraud alerts, self-serve appeal & dispute flow.	■499–■1,999/mo flat SaaS or volume-metered
3. Established Merchant	Multi-location retailers, scaling D2C brands	Full case-investigation dashboard, instant-settlement eligibility via trust tier, priority appeal handling, own-account network visibility.	■5,000–■25,000/mo SaaS or 0.02%–0.05% bps on instant settlement volume

4. Enterprise / Large Platform	Large retailers, big marketplaces with sub-sellers	Dedicated analyst console, custom SLAs, API access to trust signals for sub-merchant vetting, white-labeled trust badge.	Annual license + protected volume bps (negotiated)
5. Aggregator / Bank (B2B2B)	Other payment aggregators, acquiring banks, neobanks	Full engine licensed as Risk-as-a-Service, raw API integration, cross-platform syndicate surveillance.	Per-query or annual API licensing (highest margin)

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- 1. Starter (Individual & Informal Sellers): Public trust badge only. Free or nominal flat fee (■49–■99/month). Serves as a network-effect funnel.
- 2. Growing SMB (Small D2C & Retailers): Basic risk dashboard, trust badge, limited alerts, self-serve appeal flow. ■499–■1,999/month flat SaaS.
- 3. Established Merchant (Multi-Location & Scaling D2C): Full investigation dashboard, instant-settlement eligibility, priority appeals, network visibility. ■5,000–■25,000/month or 0.02%–0.05% bps on instant payouts.
- 4. Enterprise / Large Platform (Marketplaces & Retailers): Dedicated analyst console, custom SLAs, sub-merchant vetting API, white-labeled badge. Annual contract + volume bps.
- 5. Aggregator / Bank (B2B2B): Full engine licensed as Risk-as-a-Service, API-first cross-platform surveillance. Per-query or annual enterprise licensing.
- Projected Loss Mitigation: In out-of-sample benchmark evaluation on 321 unseen holdout merchants (from 1,070 corpus), extrapolated fraud-loss prevention is estimated at approximately ■5.90 Crore against ■3,108.98 estimated false-positive friction cost (projected simulation, not measured in live production).
- False-Positive Appeal Recovery: Integrated appeal dispute tracking quantifies operational capital restored to verified merchants (e.g. ■28.4L recovered with a 4.2-hour resolution SLA).

Engineering Finding: Shared-Infrastructure False Positives

During initial graph pipeline testing, we identified a notable source of false-positive ring detections: unrelated merchants that used the same generic payment gateway or payment service provider (PSP) handle were occasionally linked into collusive clusters by graph projection.

Fix Implemented: We added a shared-infrastructure filtering step in the network detector that down-weights and excludes edges formed purely by generic processor or gateway infrastructure. A collusive ring candidate now requires at least one strong personal identifier (bank account, phone number, or device hardware fingerprint) or multi-identifier corroboration, preventing generic payment infrastructure from generating spurious alerts.

Results & Evaluation

Benchmark Presentation Note: Standard benchmark confirms separation of obviously anomalous behavior; the harder out-of-sample holdout (30% split, 321 test merchants from a 1,070-merchant corpus across 5 random seeds) is the primary performance result. All metrics represent controlled synthetic benchmarks.

Holdout Performance & Multi-Seed Robustness (5 Random Seeds)

Metric	Standard Split (Sanity Check)	Harder Holdout (30% Unseen)	Multi-Seed (Mean ± Std)
Precision (PPV)	100.0%	98.6%	99.2% ± 1.6%
Recall (Sensitivity)	100.0%	86.4%	89.1% ± 2.6%
F1-Score	100.0%	92.1%	0.939 ± 0.017
ROC-AUC	100.0%	95.9%	0.989 ± 0.010
Planted Ring Detection Rate	100.0%	95.1%	95.0% ± 1.4%
Mule Shell Detection Rate	100.0%	60.0%	68.2% ± 8.6%

Adversarial Robustness & Evasion Stress Testing

To evaluate model resilience against evasion-aware fraudsters, we built an adversarial synthetic batch generator where fraudulent rings specifically circumvent the top behavioral signals identified by SHAP feature importances:

• Ticket Normalization: Avoids round-denomination flags by generating realistic fractional price distributions. • Buyer Entropy Dilution: Synthesizes artificially diversified buyer pools to suppress HHI concentration alerts. • Organic Settlement Cadence: Randomizes payout timing and introduces gradual volume ramps instead of sudden bursts.

Evaluation Findings: Precision remains 83.3%, while recall drops significantly to 10.0%, and ring detection drops to 25.0% (5/20 merchants caught).

Framing: This quantifies real degradation under evasion-aware conditions our team could anticipate — not a claim of general adversarial robustness. No fraud model can claim general robustness without live production exposure.

Metric	Standard Baseline	Harder Holdout	Adversarial Evasion Stress
Precision	100.0%	97.2%	**83.3%**
Recall (Sensitivity)	100.0%	86.4%	**10.0%**
Planted Ring Detection	100.0%	95.1%	**25.0%**
Mule Shell Detection	100.0%	60.0%	**0.0%**

Score Stability & Feature Perturbation Analysis

We evaluated score stability across two distinct populations under a ±5% uniform feature perturbation:

• Non-Borderline Cohort (Risk Score <45 or >75, n=20): 0.0% tier flips observed under ±5% feature perturbation, confirming baseline classification stability far from the decision boundary. • Borderline Cohort (Risk Score 50–70 near threshold 60.0, n=16): 12.5% tier flips (2/16 merchants) observed under a ±5% feature perturbation.

Boundary Sensitivity Rationale: Boundary sensitivity is an expected mathematical property of any threshold-based classifier near its decision boundary. The 48-hour cap-and-escalate settlement hold policy is the structural safeguard against acting irreversibly on a borderline flip.

Approximated Blind Hard-Negative Evaluation

Evaluated against 10 independently designed real-world edge cases, the engine achieved 80.0% accuracy (8/10 correctly classified):

• Correctly Classified (8 Cases): Seasonal festival surge (Diwali silks), PSU bank merger migration, B2B industrial chemicals, weekend night-market food vendor, creator drop apparel burst, rare book art dealer, franchise shared WiFi subnet, and quarterly round tuition fees. • Misclassified Edge Cases (2 Cases): M-BLIND-05 (tech incubator sharing address with 7 firms) and M-BLIND-09 (corporate group travel agent with high buyer concentration).

Framing Note: Framed explicitly as an 'approximated blind check' rather than a true blind study, as the 10 test scenarios were designed in a separate pass by the same engineering team.

Ring-Size Sensitivity Breakdown (Harder Split)

Ring Size Bracket	Planted Rings (Holdout)	Detected Rings	Detection Rate
2-3 merchants	15	13	**86.7%**
4-6 merchants	15	15	**100.0%**
7-10 merchants	4	4	**100.0%**
10+ merchants	2	2	**100.0%**

Audited False-Positive Cost Model (Hand-Verifiable)

Quantified Merchant Friction Formula:
Cost = Delayed Hold Volume (INR 152,765.42) × 2.0% Friction Rate = INR 3,055.31
Across 2 false-positive merchants in the 30% holdout split, only the volume processed during the 3-day hold window (INR 1.55 Lakhs) is delayed (cumulative 60-day volume is INR 15.07 Lakhs). Reversible settlement holds keep checkout active with zero buyer cart abandonment.

False-Negative Recovery Rate Sensitivity Analysis

Scenario	Recovery Rate	Gross Loss Prevented	Net Fraud Savings (INR)
Conservative	5%	INR 65,933,222.58	**INR 62,464,888.87**
Baseline (Assumed)	15%	INR 58,992,883.36	**INR 55,889,315.80**
Optimistic	30%	INR 48,582,374.53	**INR 46,025,956.19**

Threshold Selection Methodology (Why 60.0?)

Operating threshold 60.0 was selected via a structured threshold sweep grid search on the 70% training partition (749 merchants). Threshold 60.0 corresponds to the HIGH risk tier boundary where targeted interventions occur, achieving 100.0% precision with 0 false positives and 95.8% recall on training data. This balances zero merchant friction on legitimate accounts with high fraud capture. The 30% holdout test partition remained strictly out-of-sample and verified 99.2% ± 1.6% precision, 89.1% ± 2.6% recall, and 0.939 ± 0.017 F1-Score across 5 random seeds.

Known Limitations & Engineering Roadmap

• Adversarial Evasion Sensitivity Gap (Precision 83.3%, Recall dropping to 10.0%, Ring Detection to 25.0%): When fraudsters specifically normalize ticket sizes, randomize payout intervals, and dilute buyer concentration, single-merchant behavioral ML models experience severe recall degradation. Graph ring detection catches shared hardware/accounts, but standalone behavioral signals drop significantly. Roadmap: Cross-merchant behavioral embedding similarity, hardware-rooted device attestation, and velocity trajectory graph edges.

• Blind Hard-Negative Edge Cases (80.0% Accuracy, 8/10 correctly classified): Blind testing surfaced 2 cases where legitimate edge cases received high risk scores: M-BLIND-05 (tech incubator sharing a corporate address with 7 other firms) and M-BLIND-09 (corporate group travel agent with high corporate buyer concentration). This indicates that physical address sharing filters and high-concentration B2B heuristics require deeper contextual domain conditioning.

• Sleeper Mule Detection on Low-Velocity Accounts (68.2% ± 8.6% Detection Rate): Subtle sleeper accounts deliberately mimic legitimate small merchants by processing modest ticket sizes with zero customer disputes during their incubation window. Roadmap: Longer observation windows, velocity-drift trajectory tracking, and buyer entropy evolution metrics.

• Small Syndicate Detection Sensitivity (86.7% Detection Rate on 2–3 Merchant Rings, 13/15 caught): 2-to-3 merchant rings yield sparse bipartite subgraphs with fewer shared entities than large syndicates. Roadmap: Multi-hop temporal graph analysis, fuzzy device/IP subnet correlation, and higher-order motif clustering.

Banking & Payment Aggregator Compliance Architecture

• 48-Hour Maximum Hold Cap: All settlement holds enforce an automated 48-hour hard ceiling with mandatory 24-hour supervisor escalation, eliminating indefinite fund freezes. • Proactive Merchant Notification: Every hold automatically dispatches an immediate merchant notification containing the specific operational rationale, required evidence checklist, and direct dispute submission link, logged with immutable audit event MERCHANT_NOTIFICATION_DISPATCHED. • Compliance Scope Disclaimer: These operational safeguards are internal risk engineering designs aligned with the core principles of RBI Payment Aggregator Guidelines (DPSS.CO.PD.No.1810/02.14.008/2019-20). They are designed for risk operational fairness, not certified banking compliance products.

Projected Loss Mitigation & Business Impact

Holdout Monitored Volume	Suspicious Fraud Volume (TP)	Projected Loss Prevented (85%)	Net Fraud Savings (Est.)
INR 42.1 Crore	INR 6.94 Crore	**INR 5.90 Crore**	**INR 5.59 Crore (18x ROI)**